



Stochastic Processes

Lecture 20

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An Important Result

Lemma (An Important Result)

1. For any state $y \in \mathcal{X}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k(y, y) = \begin{cases} 0, & f_{yy} < 1, \\ \frac{1}{\mu_{yy} - 1}, & f_{yy} = 1. \end{cases}$$

2. If x, y are two states such that $f_{xy} = 1$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k(x, y) = \begin{cases} 0, & f_{yy} < 1, \\ \frac{1}{\mu_{yy} - 1}, & f_{yy} = 1. \end{cases}$$

On Open and Closed Communicating Classes

Lemma (On Open and Closed Communicating Classes)

Every state in an open communicating class is transient.

If a communicating class is closed and has finitely many states, then every state in it is positive recurrent.

Corollary:

- Irreducible + finite state space $\implies$ positive recurrence

Invariant (Stationary) Distribution

Definition (Invariant Distribution)

Let $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ with TPM P .

A PMF $\pi = [\pi_x : x \in \mathcal{X}]$ on $\mathcal{X}$ is called the **invariant distribution** for P if

$$\pi = \pi P. \quad \text{(global balance equation).}$$

That is, for all $y \in \mathcal{X}$,

$$\pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y).$$

Remarks on Invariant Distribution

Invariant Distribution

$$\pi = \pi P, \quad \pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y) \quad \forall y \in \mathcal{X}.$$

- Global balance equation is a collection of $|\mathcal{X}|$ linear equations
- π is a **non-negative, left eigenvector** of P with eigenvalue 1
- From Chapman–Kolmogorov's equations,

$$\pi = \pi P^n \quad \text{for any } n \in \mathbb{N}.$$

- If $\mu = \pi$ (i.e., $X_1 \sim \pi$), then:
 - $X_n \sim \pi$ for all $n \in \mathbb{N}$
 - $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a stationary process (i.e., all FDDs are translation invariant)

Remarks on Invariant Distribution

Invariant Distribution

$$\pi = \pi P, \quad \pi_y = \sum_{x \in \mathcal{X}} \pi_x P(x, y) \quad \forall y \in \mathcal{X}.$$

- If P is irreducible and there exists a non-negative vector π satisfying $\pi = \pi P$, then $\pi_x > 0$ for all $x \in \mathcal{X}$
- Given a TPM P , there may be multiple invariant distributions for P (think of $P = I$)
- **(Important question):** Under what conditions on P does there exist a unique invariant distribution?

On Existence and Uniqueness of Invariant Distribution

Lemma (On Existence and Uniqueness of Invariant Distribution)

*An irreducible TPM P has a unique invariant distribution if and only if it is positive recurrent.
In this case, the unique invariant distribution $\pi = [\pi_x : x \in \mathcal{X}]$ is given by*

$$\pi_x = \frac{1}{\mu_{xx} - 1} > 0 \quad \forall x \in \mathcal{X}.$$

Summary of Lemma 3:

- Irreducible + positive recurrence $\implies$ unique invariant distribution
- Irreducible + unique stationary distribution $\implies$ positive recurrence

Proof of Lemma 3 (Uniqueness)

- **Uniqueness:** Suppose $\alpha = [\alpha_x : x \in \mathcal{X}]$ is any other invariant distribution satisfying $\alpha = \alpha P$. Then,

$$\alpha_y = \sum_{x \in \mathcal{X}} \alpha_x \left(\frac{1}{n} \sum_{k=1}^n P^k(x, y) \right) \quad \forall n \in \mathbb{N}, y \in \mathcal{X}.$$

- Keeping y fixed and taking limits as $n \rightarrow \infty$, we get

$$\begin{aligned} \alpha_y &= \lim_{n \rightarrow \infty} \sum_{x \in \mathcal{X}} \alpha_x \left(\frac{1}{n} \sum_{k=1}^n P^k(x, y) \right) \stackrel{(a)}{=} \sum_{x \in \mathcal{X}} \alpha_x \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k(x, y) \right) \stackrel{\text{Lemma 1}}{=} \sum_{x \in \mathcal{X}} \alpha_x \left(\frac{1}{\mu_{yy} - 1} \right) \\ &= \frac{1}{\mu_{yy} - 1} > 0. \end{aligned}$$

- If $\mathcal{X}$ is countably infinite, the equality in (a) is justified by DCT:

X_n : discrete random variable taking value $\frac{1}{n} \sum_{k=1}^n P^k(x, y)$ with probability α_x

- Observe that $|X_n| \leq 1$, $X_n \xrightarrow{\text{a.s.}} \frac{1}{\mu_{yy} - 1}$.

Example

- Consider a homogeneous DTMC with $\mathcal{X} = \{0, 1, 2, \dots\}$ and transition probabilities

$$P(0, x) = \left(\frac{1}{2}\right)^x, \quad P(x, x+1) = \frac{1}{2} = P(x, 0) \quad \forall x \neq 0.$$

1. Draw the transition graph.
2. Is the Markov chain irreducible?
3. Classify the states as transient, positive recurrent, or null recurrent.
4. Compute μ_{xx} for all $x \in \mathcal{X}$.

Example

- Determine the expected number of independent tosses of a fair coin required to observe the pattern “HTH” for the first time.

Example

- **[Goldmann Sachs Interview Question @IISc] [2017-18 Placement Season]**

Imagine that you are playing a game that involves tossing a coin of bias 0.9 repeatedly and independently, with 3 lives at the start of the game. At each round, if the coin toss results in a tail, you lose one life. However, if the coin toss results in a head, you gain back your lost lives one at a time. If you have all 3 lives and the coin toss results in a head, nothing changes, and you continue tossing. You are allowed to play until you lose all your lives.

What is the expected number of tosses until finish?

Ergodicity and Ergodic Theorem

Definition (Ergodicity)

A TPM P is said to be **ergodic** if it is irreducible, aperiodic, and positive recurrent.

Ergodicity and Convergence to Stationary Distribution

Theorem (Ergodicity and Convergence to Stationary Distribution)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a homogeneous DTMC on a finite state space $\mathcal{X}$.

Let μ_n denote the PMF of X_n for each n .

If P is ergodic with unique invariant distribution $\pi = [\pi_x : x \in \mathcal{X}]$, then

$$\lim_{n \rightarrow \infty} d_{\text{TV}}(\mu_n, \pi) = \lim_{n \rightarrow \infty} \frac{1}{2} \sum_{x \in \mathcal{X}} |\mu_n(x) - \pi_x| = 0,$$

where d_{TV} denotes the **total variation distance**.

Remarks:

- Although the proof will be presented for $\mathcal{X}$ finite, the result holds even when $\mathcal{X}$ is countably infinite
- The above theorem is a statement about convergence in distribution

- **Role of aperiodicity:**

If P is irreducible, aperiodic, and has $|\mathcal{X}| < +\infty$, then there exists $N \in \mathbb{N}$ such that $P^N > \mathbf{0}$

- The proof of ergodicity theorem is based on an important idea called **coupling**

Proof of Theorem 4 via Coupling

- Let $\mathbf{Y} = \{Y_n\}_{n \in \mathbb{N}}$ be an **independent copy** of $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$, but with one difference:

$$X_1 \sim \mu, \quad Y_1 \sim \pi.$$

- $\mathbf{X}$ and $\mathbf{Y}$ evolve according to the same TPM P
- Let τ be the first time instant at which both $\mathbf{X}$ and $\mathbf{Y}$ meet, i.e.,

$$\tau := \inf\{n \in \mathbb{N} : X_n = Y_n\}.$$

The random variable τ is called the **coupling time**

- Let $Z_n = (X_n, Y_n) \in \mathcal{X} \times \mathcal{X}$
- Exercise:** $\{Z_n\}_{n \in \mathbb{N}}$ is a homogeneous DTMC on the state space $\mathcal{X} \times \mathcal{X}$ with TPM

$$Q\left((x, w), (y, z)\right) = P(x, y) \cdot P(w, z).$$

Proof of Theorem 4 via Coupling

- By Chapman–Kolmogorov,

$$Q^n((x, w), (y, z)) = P^n(x, y) \cdot P^n(w, z) \quad \forall n \in \mathbb{N}.$$

- Because P is irreducible and aperiodic, there exists $N \in \mathbb{N}$ such that every entry of P^N is strictly positive. As a result, every entry of Q^N is strictly positive, and thus Q is irreducible.
- From Question 4(d) of Homework 04, it follows that $\lambda = 1$ is a simple eigenvalue of Q^N .
- Thus, there exists a unique eigenvector α^* such that $\alpha^* = \alpha^* Q^N$.
- α^* must then be the unique eigenvector satisfying $\alpha^* = \alpha^* Q$.
- By brute-force calculations, one can verify that

$$\alpha(x, y) = \pi(x) \cdot \pi(y), \quad x, y \in \mathcal{X},$$

satisfies $\alpha = \alpha Q$

- Thus, $\alpha^*(x, y) = \pi(x) \pi(y)$ is the unique invariant distribution for Q .
- Q is irreducible and has a unique stationary distribution α^* .
From Lemma 3, it then follows that Q is positive recurrent.

Proof of Theorem 4 via Coupling

- Q positive recurrent implies that starting from any (X_1, Y_1) , a state of the form (x, x) will be visited infinitely many times for each $x \in \mathcal{X}$

This, in turn, implies that $\mathbb{P}(\{\tau < +\infty\}) = 1$

- The following is true for the coupled processes:

$$\mathbb{P}(\{X_n = y, n \geq \tau\} \mid \{X_\tau = x\}) = \mathbb{P}(\{Y_n = y, n \geq \tau\} \mid \{Y_\tau = x\}).$$

- We also have

$$\begin{aligned}\mathbb{P}(\{X_n = y, n \geq \tau\}) &= \sum_{x \in \mathcal{X}} \mathbb{P}(\{X_n = y, n \geq \tau\}) \cdot \mathbb{P}(\{X_\tau = x\}) \\ &= \sum_{x \in \mathcal{X}} \mathbb{P}(\{Y_n = y, n \geq \tau\}) \cdot \mathbb{P}(\{Y_\tau = x\}) \\ &= \mathbb{P}(\{Y_n = y, n \geq \tau\}).\end{aligned}$$

Proof of Theorem 4 via Coupling

- Finally, we note that

$$\begin{aligned} |\mu_n(y) - \pi(y)| &= \left| \mathbb{P}(\{X_n = y\}) - \mathbb{P}(\{Y_n = y\}) \right| \\ &= \left| \mathbb{P}(X_n = y, \tau > n) - \mathbb{P}(Y_n = y, \tau > n) \right| \\ &\leq 2 \mathbb{P}(\{\tau > n\}) \end{aligned}$$

- Taking limits as $n \rightarrow \infty$, and noting that $\mathbb{P}(\tau < +\infty) = 1$, we arrive at the result